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Description This document relates to Signa Horizon LX systems with 8645/GRAM hardware. This test is the primary means for functionally testing the GRAM Digital Tuning Board. This test uncovers other problems in the Gradient Driver subsystem, but focuses on the Digital Tuning Board.

1- INTRODUCTION

The Digital Tuning Board Test is invoked via the Diagnostics Menu found on the service page. This button is available only if the `MRconfig.cfg` file indicates `digiTuningBoard = "yes"`. The test produces an error and does not execute if Digital Tuning Boards are not present on ALL axes.

A GRAM must be present for Digital Tuning Boards to be present. Although ASM boards are not required, each axis must have a MIF, GRAM, and GASM.

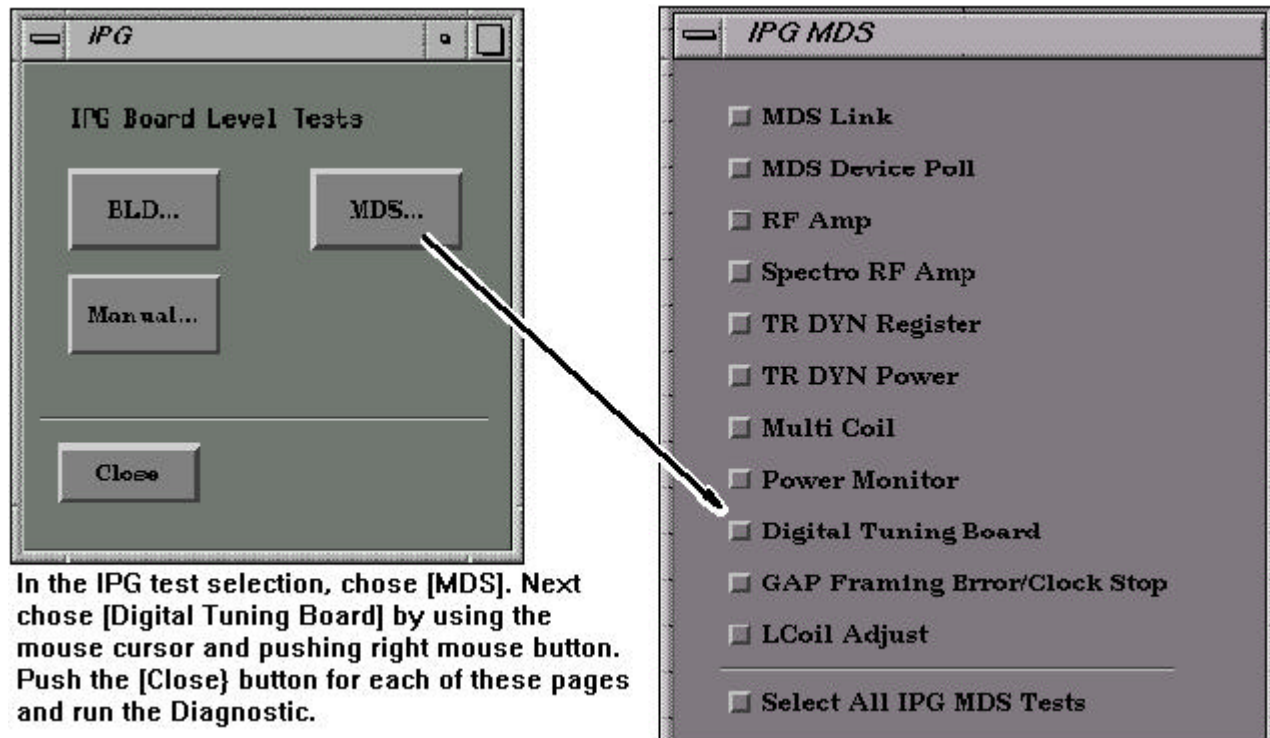
Normal calibration (GRAM Tuning, Grafidy, etc.) must be performed prior to executing this test.

Note

DACi ramping speed - Due to the DACi ramping speed used in this test, GRAM Overflow and Underflow overloads occur during the execution of the test. This is reflected by a flashing amber Overload LED on each GRAM Control Board. These overloads are completely normal and should be ignored. They do not cause damage to the hardware since the gradients are in Standby for this test. Software ignores these overloads and does not report them.

2- INVOKING THE DIGITAL TUNING BOARD TEST

1. When Signa Horizon LX software is up, go to the service desktop and start the service diagnostics.
2. Open the IPG group by pushing the box with the mouse cursor and left mouse button below the IPG label. Follow the instructions as shown in Illustration L001.



DIAGNOSTIC MAIN MENU
ILLUSTRATION L001

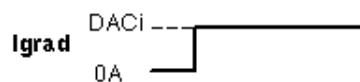
3- DIAGNOSTIC TEST RESULTS

Once the diagnostics have completed, data will appear indicating the faults that have occurred, if any. If a fault has occurred, remember to use the system error log to identify a recommended service strategy.

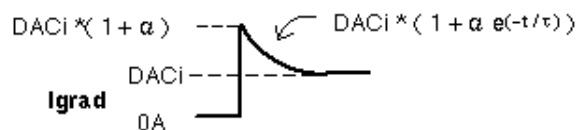
4- TEST DESCRIPTION

By cycling through a series of Digital Tuning Board time constants (t) and gains (a), the exponential response of the GRAM DACi, or Igrad signal, is examined. Illustration L4041A shows the Exponential Response of the Igrad signal with and without eddy current compensation.

Without Eddy Current Compensation
($\tau = 0, \alpha = 0$):



With Eddy Current Compensation:



L4041A

EXPONENTIAL RESPONSE FOR ECC
ILLUSTRATION L4041A

The overall time constant (t) is actually composed of the summation of seven time constants. Together, these time constants provide greater resolution and overall range. Likewise, the overall gain (α) is actually composed of the summation of seven gains. Together, these gains also provide greater resolution and range.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	May 22, 1998	J. Saperstein	Initial Conversion from Toolbook to Word format.